



DEALER BEST PRACTICE SERIES

3500 Crankshaft and Camshaft Storage

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC
Management

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1.0 Introduction

Trakindo's Component Rebuild Centers (CRCs) have a large amount of Work in Process (WIP) when providing Component Rebuild engines. Two of the larger components in the CRCs are crankshafts and camshafts waiting to be installed. The current storage method for these specific components is inefficient, which affects the turnaround time. The CRCs end up with valuable space being taken up by WIP, resulting in the area looking congested, and exposing the components to potential damage.

When considering contamination control and 5S, the components (3500 series crankshaft and camshafts or smaller) when placed on the floor using the current method take up valuable space and creates a potential safety hazard. In some instances when the parts are small enough they are placed on desk space in the CRCs. The current method makes it difficult for the technicians to retrieve the component when needed. As a result, Trakindo needed to implement a new method to allow crankshafts and camshafts to be stored efficiently in a clean and tidy environment, which makes possible safe and easy staging and transport.

2.0 Best Practice Description

This Best Practice is for storage and staging management of the 3500 Series Crankshafts and Camshafts. It can be also applied to other models. The focus should be on the staging of the shafts for contamination control, easy access, maximum floor space utilization, component quality (shafts not bending), and technician safety. Examples from Trakindo have been presented as well as some of the best in class storage methods gathered from around the world.

Trakindo Crankshaft Storage

The new method that has been implemented is to store the crankshafts in the vertical position using racks and a lifting bracket that is bolted onto the flywheel bolt holes in the crankshaft. Plastic wrap is applied to the components to provide protection while in storage. This provides a safe and efficient method of storing, lifting, and moving the crankshafts.

The lifting bracket is vital so the shaft can be removed vertically without moving the other shafts in the rack. The rack gage dimension must be wide enough to allow clearance for vertical lifting of the shaft.

Trakindo created dedicated stands that can store 10 crankshafts each. A step, ladder, or crane hook guiding pole is required for access to the stand to lift the crankshaft into place.

At least two standing racks are needed - one for the crankshafts that have been just remove from engine and waiting for rebuild, and another one for the crankshafts that have been recondition and waiting to be installed.

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Figures 1 and 2: Trakindo's crankshaft storage and staging process before implementation of storage racks.



Figures 3 and 4: Trakindo's crankshaft storage and staging process after implementation of storage racks.

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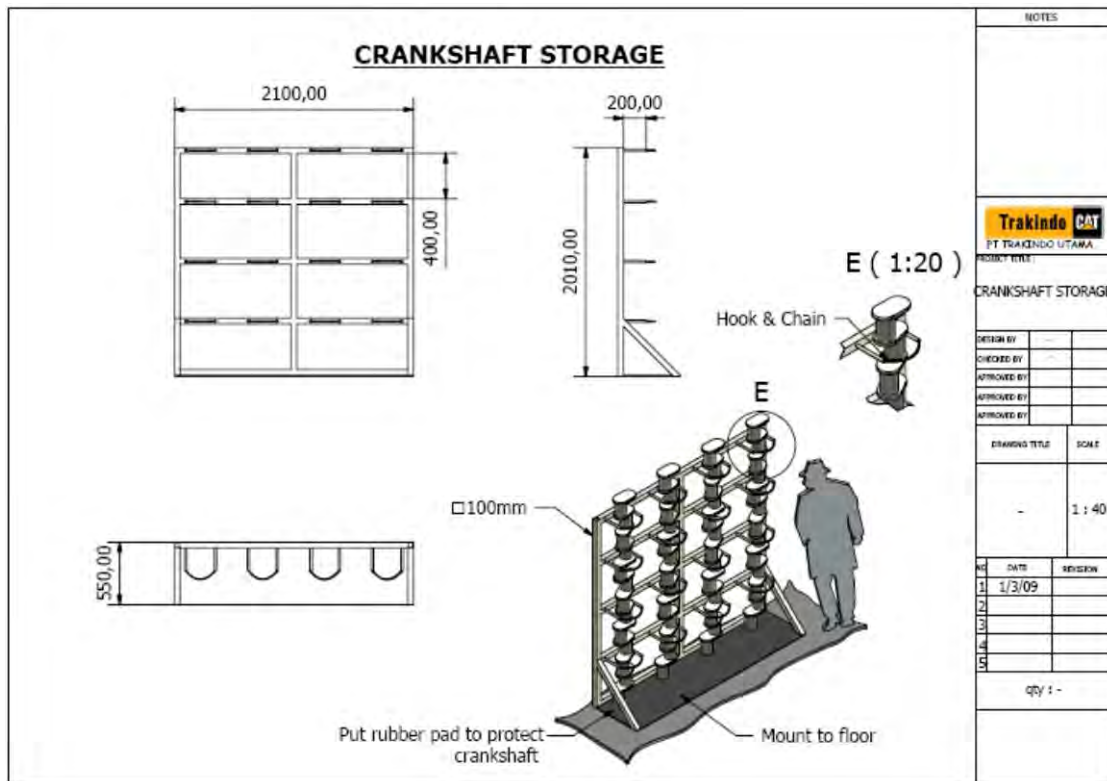


Figure 5: Trakindo crankshaft rack drawing.

Trakindo Camshaft Storage

The new method that has been implemented to store the camshafts is to develop racks to hold the components in a vertical position. Plastic wrap is applied to the components to provide protection while in storage.

At least two standing rack are needed - one for the camshafts that have been just remove from engine and waiting for rebuild, and another one for the camshafts that have been recondition and waiting to be installed.

For 3500 engine series camshaft or bigger, Trakindo uses a webbing sling to lift the camshaft to avoid the scratching and other damage when transporting.

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Figure 6 and 7: Trakindo's camshaft storage and staging process before implementation of storage racks.



Figures 8 and 9: Trakindo's camshaft storage and staging process after implementation of storage racks.

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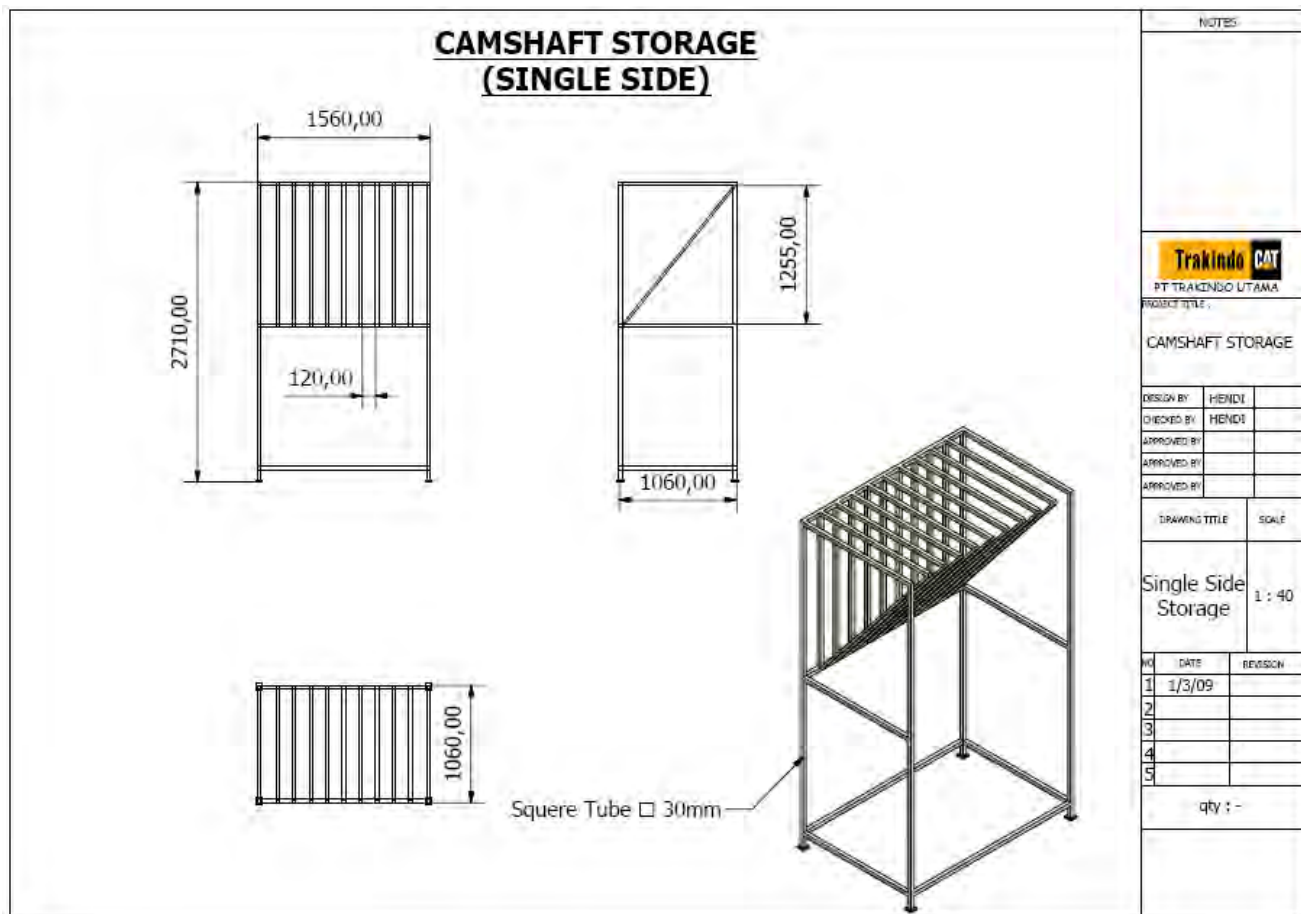


Figure 10: Trakindo camshaft rack drawing.

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Additional Best in Class Examples for Crankshaft and Camshaft Storage

Ferreyros has implemented a reusable canvas cloth with Velcro straps to protect their camshafts while in vertical rack storage instead of traditional plastic shrink wrap. This provides an easy method to remove and reuse protection for the components.



Figure 11: Ferreyros camshaft storage using canvas cloth with Velcro straps.

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Gecolsa and Finning Chile have also implemented racks to provide easy access to the components both vertically and horizontally.



Figure 12 and 13: Gecolsa crankshaft storage and staging racks.



Figure 14 and 15: Finning Chile crankshaft storage and staging racks.

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3.0 Implementation Step

- Determine area in shop to be designated for storage and staging rack.
- Designate *Incoming* and *Outgoing* racks in rebuild/processing work bay.
- Assure the area has adequate space and access for equipment to move the component with a crane – no lift trucks. Best to be able to move components vertically and horizontally.
- Install the layout/equipment changes on the shop floor.
- Promote change feedback as installation may make new problems surface.
- Train employees to use the new procedures/equipment.
- Review process/equipment performance once established.
- Establish and install any adjustments as needed.

4.0 Benefits

Safety

Well designed/organized standing rack will provide a safer and cleaner working environment.

Organization

Components are consistently staged for incoming and outgoing status. Floor space is maximized by the components taking up minimal storage space.

Quality

Simplified staging and tidy WIP areas promotes technician professionalism and higher standards, while providing shaft protection, contamination control, and shaft cleanliness resulting in increased brand image.

5.0 Resources Required

Facility

Fabrication material.

Reorganized work area to allow for vertical storage area with incoming and outgoing staging.

People

Establish the new process, layout, and equipment improvement through the team.
Remove replaced items from operation to prevent less effective storage system.

6.0 Supporting Attachment/Reference

None

7.0 Related Best Practices

- 1106-4.4-1038 - In-Process Staging Strategy
- 1006-4.4-1032 - Piston, Rod, and Liner Staging Equipment

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